AI Powered Medical Imaging Assistant

A major project report submitted in partial fulfillment of the requirement for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

Submitted by

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Under the guidance & supervision of

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December 2025

Candidate's Declaration

We hereby declare that the work presented in this major project report entitled **'AI Powered Medical Imaging Assistant'**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is an authentic record of our own work carried out during the period from July 2025 to December 2025 under the supervision of **Mr. Faisal Firdous.**

We further declare that the matter embodied in this report has not been submitted for the award of any other degree or diploma at any other university or institution.

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Date: 10/12/2025
Place: JUIT Solan

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Supervisor’s Certificate

This is to certify that the major project report entitled **‘AI Powered Medical Imaging Assistant’**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is a bonafide project work carried out under my supervision during the period from July 2025 to December 2025.

I have personally supervised the research work and confirm that it meets the standards required for submission. The project work has been conducted in accordance with ethical guidelines, and the matter embodied in the report has not been submitted elsewhere for the award of any other degree or diploma.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
U-Net	U-shaped Convolutional Network for Image Segmentation
MRI	Magnetic Resonance Imaging
CT	Computed Tomography
X-ray	X-radiation Imaging
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
GAN	Generative Adversarial Network
DDPM	Denoising Diffusion Probabilistic Model
DSC	Dice Similarity Coefficient
IoU	Intersection over Union
MSE	Mean Squared Error
STS	Standardized Test Set / Specific Dataset Name
AdamW	Adaptive Moment Estimation with Weight Decay
LR	Learning Rate
HR	High Resolution
PSF	Point Spread Function
2D / 3D	Two-/Three-Dimensional
GPU	Graphics Processing Unit
CPU	Central Processing Unit
API	Application Programming Interface

UI / UX	User Interface / User Experience
BT	Brain Tumor
MedSAM	Medical Segmentation Assisted Module
VLM	Vision-Language Model
FOV	Field of View
ReLU	Rectified Linear Unit
IoT	Internet of Things
HL	Hidden Layer (Neural Network)

ABSTRACT

Early and accurate diagnosis of brain tumors is critical for effective treatment. Medical professionals often rely on MRI scans for the detection and classification of brain tumors. However, if you have to manually analyze thousands of MRI scans, you will spend a long time doing so and the chances that your results will be misclassified or altered due to human error are high. Also, due to the nature of MRI imaging, there is often noise and poor quality on scans from different MRI machines; thus making the diagnosis even more challenging. Therefore, there is a strong need for intelligent systems to automatically identify and classify brain tumors of different types. In order to solve this problem, this project has developed an AI-Powered Brain Tumor Classification System that combines image enhancement with reliable and accurate brain tumor detection. The system employs a multi-stage deep learning pipeline to deliver its classification solutions: first, U-Net and DnCNN models are used to generate MRI images with reduced noise and background artifacts, which results in improved clarity of the images. Then, an EfficientNetB4 neural network is used to classify the enhanced MRI images into four categories: glioma, meningioma, pituitary tumor or no tumor. The entire solution is packaged in a web application that is easy to use and requires no prior technical expertise on the part of healthcare providers. Users of this application simply upload an MRI scan and receive detailed diagnostic information regarding the presence and type of a brain tumor within seconds.

Using a highly curated dataset of 7,023 brain MRI files, the system learned using Artificial Intelligence (AI) and produced an exemplary accuracy rate of 98.17% with a macro AUC result of 99.93%. The Image Enhancement process provided better quality scans using over 5 decibels in PSNR measures while achieving balanced results among all tumour classes, with greater than 98% specificity for each class. Five models were tested in the classification stage: Custom CNN, VGG16, ResNet50, Xception, and EfficientNetB4. EfficientNetB4 outperformed all others across key evaluation metrics and was chosen as the final model.